GradientCIO

PORTFOLIO ATTRIBUTION REPORT

Northstar Total Fund

Historical and forward-looking attribution · prepared for the Investment Committee

HISTORICAL PERIOD
2025-07-31 to 2026-06-30

FORWARD VIEW AS OF
2026-10-08

BASE CURRENCY
USD

BENCHMARK ROLE
Policy

01 Executive Attribution Summary

Realized and assumption-based effects

Historical and forward views reconcile

ATTRIBUTION CONCLUSION

The portfolio's realized active return was -0.6pp, led by a -0.8pp public-equity selection effect S4. Governed ex ante attribution is +0.4pp on persisted one-year expected returns, with fixed-income selection the largest positive effect S5. Both lanes reconcile within their returned tolerance, but their realized linked and single-period expected bases are not interchangeable S4 S5.

HISTORICAL ACTIVE RETURN	LARGEST HISTORICAL EFFECT	EX ANTE ACTIVE RETURN	LARGEST SUB-ALLOCATION EFFECT
-0.6pp	-0.8pp	+0.4pp	-0.4pp
12 months to 2026-06-30 S4	Public equity selection S4	One-year expected basis S5	US equity cumulative selection S4

AT A GLANCE

AREA	EVIDENCE-BASED FINDING	STATUS
Historical result	Active return was -0.6pp; public-equity selection was the largest negative effect S4.	Available
Ex ante result	Expected active return is +0.4pp; fixed-income selection is the largest positive effect S5.	Available
Reconciliation	Both total effects reconcile to their returned active return within tolerance S4 S5.	Passed
Assumptions	Ex ante effects use persisted base-currency expected returns and current cohort weights S5.	Assumption
Coverage	All required sample sources are available S1 S2 S3 S4 S5.	Complete

Scope

This is a monitoring analysis, not an investment recommendation or forecast. Ex ante expected returns and effects are assumptions, not guarantees.

Key judgment — Reconciliation

Observation: The two attribution lanes, historical and ex ante, both reconcile S1. Why it matters: Both lanes give one consistent explanation of active return. Uncertainty: Ex ante effects depend on capital market assumptions. What would change the view: A refreshed assumption set that moves either lane by 0.1pp.

Key judgment — Active return

Observation: Historical active return was -0.6pp over the period S2. Why it matters: It is the shortfall that the attribution effects explain. Uncertainty: Benchmark returns use the policy weights in force each month. What would change the view: A restated benchmark history that moves active return by 0.1pp or more.

Key judgment — monitoring priority

Observation: US equity cumulative selection is the largest sub-allocation effect at -0.4pp S4. Why it matters: The depth-1 result identifies where the root detractor is concentrated. Uncertainty: The rerun covers only returned child allocations. What would change the view: A refreshed depth-1 result identifying another leading sub-allocation.

02 Benchmark and Data Basis

The two attribution lanes use different evidence

Portfolio	**Northstar Total Fund** S2
Period	**2025-07-31 to 2026-06-30** S4
Parent cohort	**Root** S2
Benchmark role	**Policy** S4
Base currency	**USD** S2
Historical basis	**Monthly total returns; beginning-of-period weights** S4
Ex ante basis	**One-year expected returns; current normalized cohort weights** S5
Data scope	**Illustrative sample data** S1

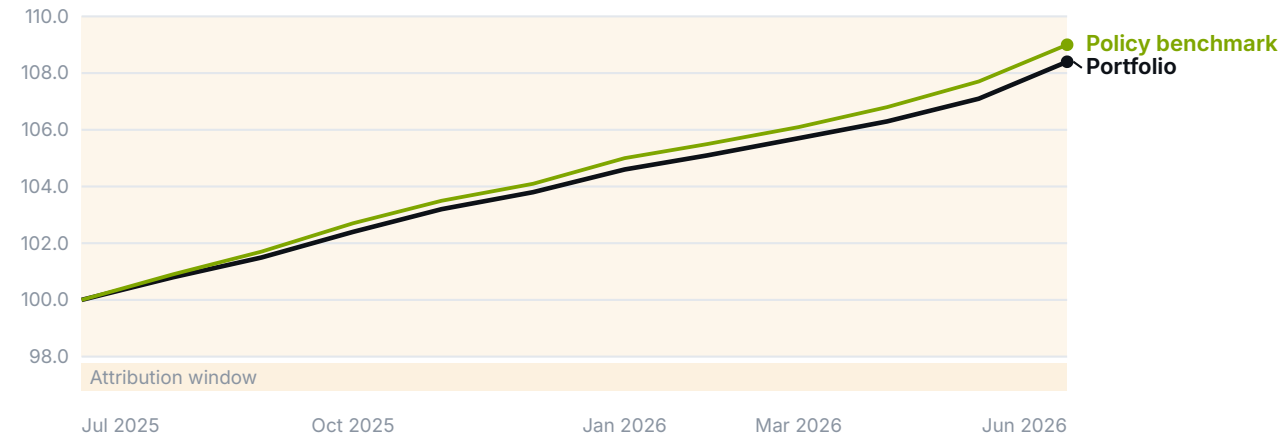
METHOD BOUNDARY

LANE	WEIGHTS	RETURNS	HORIZON / LINKING	METHOD VERSION	SOURCE
Historical	Beginning of period	Realized total return	12 monthly periods / symmetric Carino	realized_brinson_fachler	S4
Ex ante	Current actual and persisted policy weights, normalized	Annualized expected return	One expected period / no linker	1.0.0	S5

03 Historical Returns Context

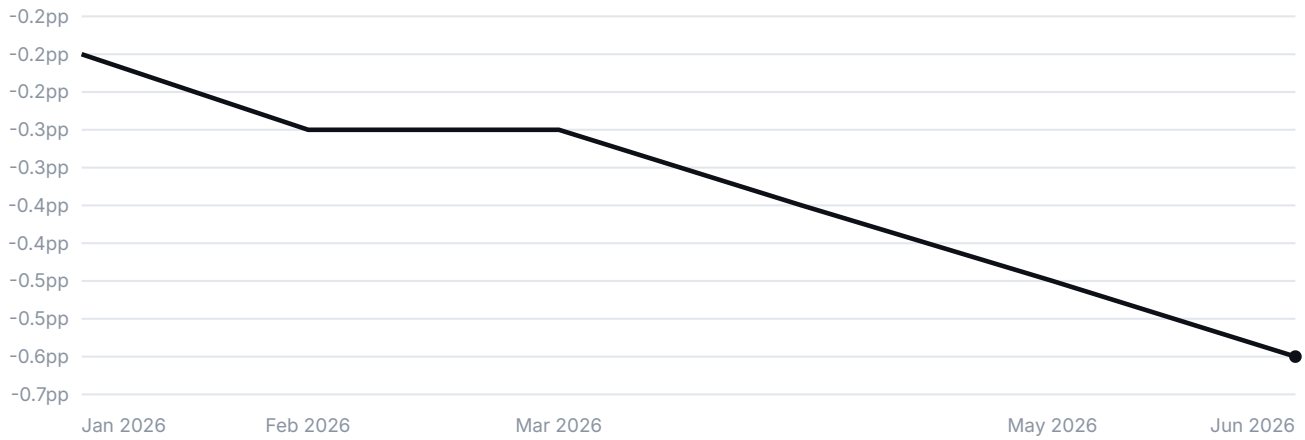
Observed performance provides context, not causal attribution

GROWTH OF 100



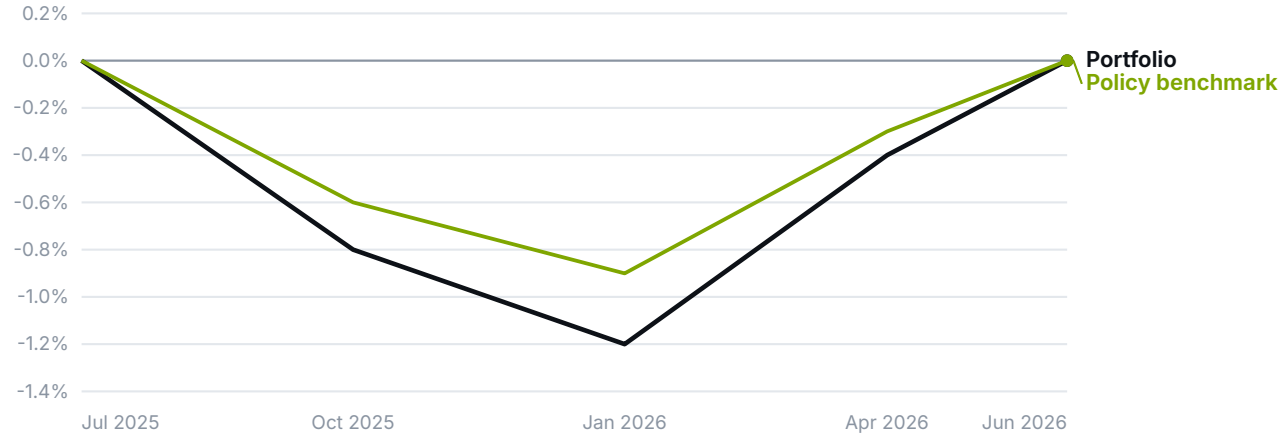
Full returned monthly series on the policy-benchmark entity basis S3.

ROLLING 12-MONTH EXCESS



Returned rolling 12-month arithmetic excess S3.

DRAWDOWN



Returned drawdown series S3.

STANDARD PERIODS TO 2026-06-30

PERIOD	PORTFOLIO	BENCHMARK	EXCESS	COVERAGE
QTD	2.8%	3.2%	-0.4pp	Complete
YTD	4.6%	5.1%	-0.5pp	Complete
1Y	8.4%	9.0%	-0.6pp	Complete
3Y (ann.)	7.1%	7.5%	-0.4pp	Complete

Returned labels, annualization and coverage are preserved S3 .

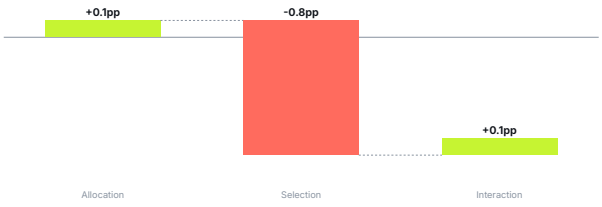
Analysis — Relative returns trailed recently

Observation: The portfolio trailed over all three periods: QTD, YTD and 1Y S3 . Why it matters: The pattern establishes context for attribution results. Uncertainty: Aggregate return coverage does not identify causal effects. What would change the view: A later full-period report showing sustained positive excess return.

04 Historical Attribution

Realized Brinson-Fachler effects with symmetric-Carino linking

ALLOCATION → SELECTION → INTERACTION



Returned linked effects S4 .

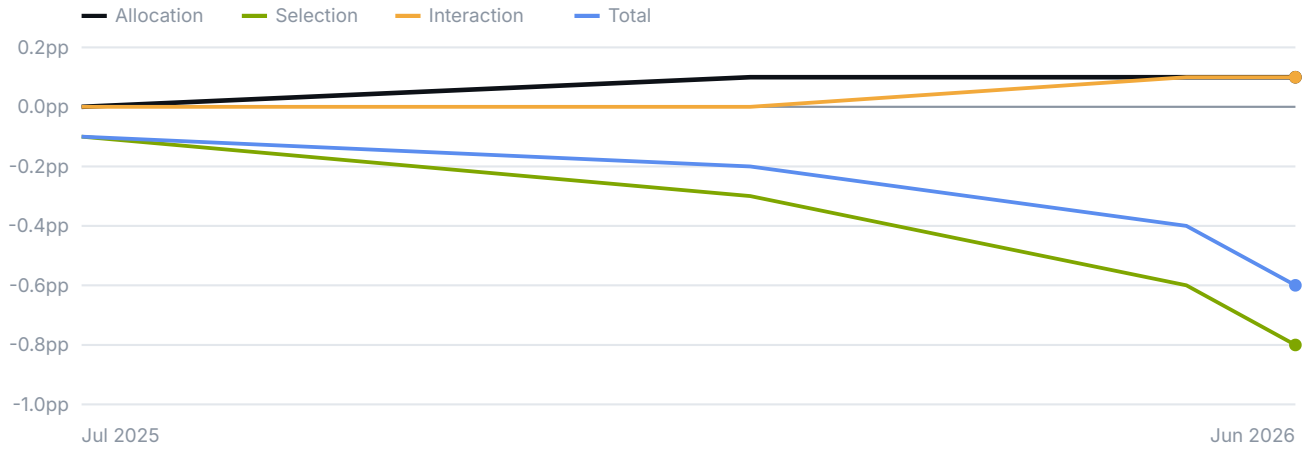
HISTORICAL RECONCILIATION AND METHOD

Active return	**-0.6pp** S4
Total effects	**-0.6pp** S4
Method	**Realized Brinson-Fachler** S4
Linking	**Symmetric Carino** S4
Currency	**USD** S4

HISTORICAL SEGMENT TOTAL EFFECTS



CUMULATIVE LINKED EFFECTS



Returned cumulative symmetric-Carino linked effects S4.

SEGMENT × MONTH LINKED TOTAL EFFECT

	JUL 2025	JAN 2026	MAY 2026	JUN 2026
PUBLIC EQUITY	-0.1	-0.2	-0.2	-0.1
FIXED INCOME	0.0	0.1	0.1	0.1
PRIVATE ASSETS	0.0	-0.1	-0.3	0.0

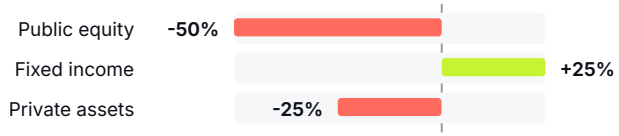
Returned monthly linked total effects by segment S4.

CONTRIBUTION TO TRACKING ERROR



Returned active-risk contribution shares S4.

RETURN EFFECT SHARE



Returned total effect divided by the sum of absolute returned segment effects S4.

CUMULATIVE SELECTION INSIDE US EQUITY



Returned cumulative selection from the depth-1 attribution rerun S4.

HISTORICAL TOTAL EFFECTS

EFFECT	CONTRIBUTION	SOURCE
Allocation	+0.1pp	S4
Selection	-0.8pp	S4
Interaction	+0.1pp	S4
Total attribution	-0.6pp	S4

HISTORICAL SEGMENT EFFECTS

SEGMENT	ALLOCATION	SELECTION	INTERACTION	TOTAL
Public equity	+0.1pp	-0.8pp	+0.1pp	-0.6pp
Fixed income	+0.1pp	+0.2pp	0.0pp	+0.3pp
Private assets	-0.1pp	-0.2pp	0.0pp	-0.3pp

Rows are returned effects and are not recalculated S4.

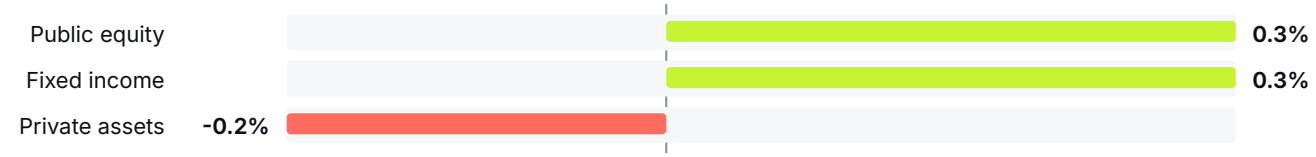
Analysis — Equity selection drove detractor

Observation: Public-equity selection is the largest detractor, and resolved managers have one open finding S4 S6. Why it matters: Effect and diligence evidence concentrate oversight there. Uncertainty: The balanced-regime model is contextual and not causal S7. What would change the view: Closed findings or refreshed results identifying another leading effect.

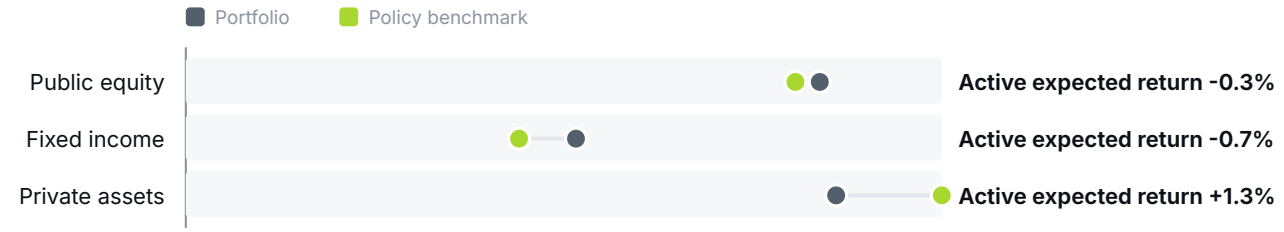
05 Governed Ex Ante Attribution

Single-period expected effects from persisted assumptions

GOVERNED EX ANTE SEGMENT TOTAL EFFECTS



EXPECTED RETURN BY SEGMENT



Returned portfolio and policy-benchmark expected returns S5.

Assumptions, not forecasts

The one-year expected returns and resulting effects are assumptions, not forecasts or guarantees S5.

EX ANTE TOTAL EFFECTS

EFFECT	CONTRIBUTION	SOURCE
Allocation	+0.1pp	S5
Selection	+0.2pp	S5
Interaction	+0.1pp	S5
Total attribution	+0.4pp	S5

EX ANTE RECONCILIATION AND METHOD

Expected active return	**+0.4pp** S5
Total effects	**+0.4pp** S5
Method / version	**Expected Brinson-Fachler / 1.0.0** S5
Horizon / linking	**One year / none** S5
Return sources	**Allocation and benchmark expected return base** S5
Assumption set / regime	**Not linked / Not linked** S5
Currency	**USD** S5

EX ANTE SEGMENT EFFECTS

SEGMENT	PORTFOLIO WT.	BENCHMARK WT.	PORTFOLIO EXP. RETURN	BENCHMARK EXP. RETURN	ALLOCATION	SELECTION	INTERACTION	TOTAL
Public equity	55.0%	50.0%	7.8%	7.5%	+0.1pp	+0.2pp	0.0pp	+0.3pp
Fixed income	30.0%	35.0%	4.8%	4.1%	+0.1pp	+0.2pp	0.0pp	+0.3pp
Private assets	15.0%	15.0%	8.0%	9.3%	0.0pp	-0.2pp	0.0pp	-0.2pp

Inputs and effects are returned by the governed ex ante contract S5.

Analysis — Assumption lineage remains incomplete

Observation: Ex ante active return is +0.4pp, but no assumption set or regime is linked S5. Why it matters: Persisted inputs cannot be tied to a named release. Uncertainty: Input dates may differ by segment. What would change the view: A named release and regime record linked to every segment.

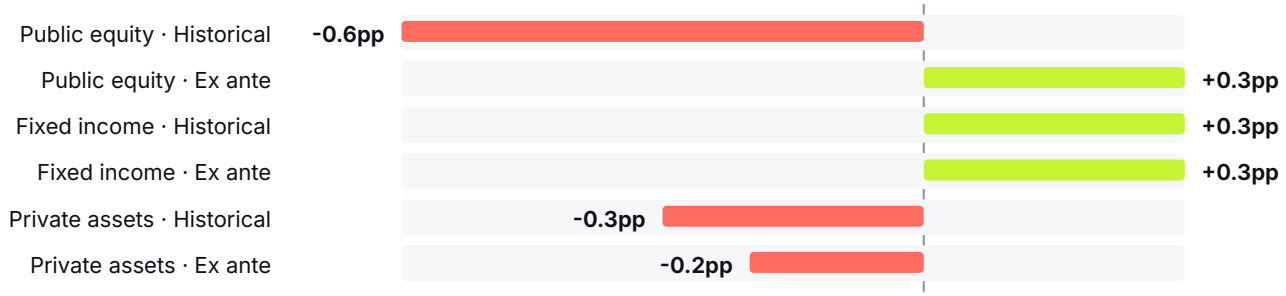
06 Historical-versus-Ex-Ante Scope and Comparison

Directional comparison with explicit basis separation

Basis difference

Historical effects use realized monthly observations and symmetric-Carino linking; ex ante effects use current normalized cohort weights and one expected period. The values are not directly interchangeable S4 S5.

HISTORICAL VS EX ANTE SEGMENT EFFECTS



Paired returned effects; lanes retain different bases S4 S5.

PUBLIC EQUITY
Changed sign
Historical − · ex ante + S4 S5

FIXED INCOME
Same sign
Historical + · ex ante + S4 S5

PRIVATE ASSETS
Same sign
Historical − · ex ante − S4 S5

DIRECTIONAL COMPARISON

SEGMENT	HISTORICAL TOTAL EFFECT	EX ANTE TOTAL EFFECT	DIRECTION	BASIS NOTE
Public equity	-0.6pp	+0.3pp	**Changed sign**	Realized linked vs expected single period S4 S5
Fixed income	+0.3pp	+0.3pp	**Same sign**	Realized linked vs expected single period S4 S5
Private assets	-0.3pp	-0.2pp	**Same sign**	Realized linked vs expected single period S4 S5

Analysis — Public equity changes sign

Observation: Public equity is -0.6pp historically and +0.3pp ex ante S4 S5. Why it matters: The sign change separates experience from expectations. Uncertainty: The lanes use different periods and weight bases. What would change the view: Refreshed persisted results on the intended policy basis.

07 Diagnostics and Limitations

Reconciliation, coverage and model boundaries

HISTORICAL
0.0 bp
Tolerance 0.000001 bp · ✓ S4

EX ANTE
0.0 bp
Tolerance 0.000001 bp · ✓ S5

RECONCILIATION DIAGNOSTICS

LANE	COVERAGE	RESIDUAL	TOLERANCE	WITHIN TOLERANCE	WARNINGS
Historical	Available	0.0bp	0.00000001pp	**Yes**	None S4
Ex ante	Available	0.0bp	0.00000001pp	**Yes**	Assumption set and regime IDs are null S5

- Historical effects cover 12 complete monthly observations and use returned symmetric-Carino linking S4.
- Ex ante effects use current actual and persisted policy benchmark weights normalized within the root sibling cohort S5.
- Persisted expected returns are in USD; currency effects are not separately included S5.
- Assumption-set and regime identities are not attached to the persisted expectations S5.
- Missing evidence would remain unavailable rather than being treated as zero.

08 Coverage

Availability of every expected source

Capabilities	**Available**	Portfolio Analytics live sample access S1
Portfolio and structure	**Available**	Root cohort; three segments S2
Historical returns context	**Available**	Complete through 2026-06-30 S3
Historical attribution	**Available**	12 monthly periods; governed result available S4
Governed ex ante attribution	**Available**	One-year expected basis; governed result available S5
Largest-detractor manager findings	**Available**	One open finding for resolved public-equity managers S6
Macro regime	**Available**	Balanced regime context for the historical period S7

A Appendix — Sources and methods

Based on 5 of 5 evidence areas.

TAG	EVIDENCE	SOURCE	AS OF	CHECK
S1	Capabilities	get_gradient_capabilities (tool metadata)	2026-10-08	**Passed**
S2	Portfolio and root cohort	list_portfolios; get_portfolio_structure	2026-10-08	**Passed**
S3	Historical returns context	get_portfolio_historical_returns	2026-06-30	**Passed**
S4	Historical attribution	get_portfolio_attribution	2026-06-30	**Passed**
S5	Governed ex ante attribution	get_portfolio_ex_ante_attribution	2026-10-08	**Passed**
S6	Largest-detractor manager findings	get_gradient_capabilities	2026-06-30	**Passed**
S7	Macro regime	get_macro_signals	2026-06-30	**Passed**

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

SOURCE AUDIT TRAIL

TAG	KEY PARAMETERS	DIGEST
S1	detail=full	Not applicable — tool metadata
S2	view=allocation_tree	sha256:2222222222222222222222222222222222222222222222222222222222222222
S3	five sections	sha256:3333333333333333333333333333333333333333333333333333333333333333
S4	policy; root; all sections	sha256:4444444444444444444444444444444444444444444444444444444444444444
S5	policy; root; all sections	sha256:5555555555555555555555555555555555555555555555555555555555555555
S6	view=open; public-equity managers	sha256:6666666666666666666666666666666666666666666666666666666666666666
S7	view=gradient_signal; regime_state	sha256:7777777777777777777777777777777777777777777777777777777777777777

Data scope for every source: Illustrative sample data. Each digest is the SHA-256 fingerprint of the exact data the report used.

B Appendix — Server Metric Methods and Disclosures

SERVER METRIC METHODS

LANE	QUANTITY	METHOD / FORMULA	VERSION	BASIS / UNITS / TOLERANCE
Historical	Allocation, selection, interaction	Realized Brinson-Fachler, Carino-linked	governed-portfolio-attribution-v1	Monthly; decimal; 1e-10 S4
Ex ante	Allocation, selection, interaction	Expected Brinson-Fachler, one period	1.0.0	Annualized; decimal; 1e-10 S5
Historical returns	Growth, rolling excess, drawdown	Indexed growth, rolling excess, peak-to-trough drawdown	governed-historical-returns-v1	Monthly; index and percent S3
Historical active risk	Tracking-error contribution	Segment share of total active risk	governed-portfolio-attribution-v1	Annualized; share S4
Historical periods	Linked monthly and cumulative effects	Carino-linked period and segment effects	governed-portfolio-attribution-v1	Percentage points S4
Historical drill-down	Largest sub-allocation selection	Depth-1 period selection effects, as returned	governed-portfolio-attribution-v1	Cumulative percentage points S4
Ex ante assumptions	Expected return by segment	Expected returns on actual and policy weights	1.0.0	Annualized; percent S5
Comparison	Segment effects and sign agreement	Historical and ex ante effects side by side	report-contract-v1	Percentage points; not rebased S4 S5

- Ex ante returns and effects are assumptions, not forecasts or guarantees; historical attribution is realized, so the two bases are not interchangeable.
- Missing evidence is not zero.
- AI-assisted monitoring analysis; not investment, legal or compliance advice.